

Cep146 - Lab2

Group Members: Mohamed ali, Wen Xin Gao, Aidan Pacheco

1. Select a scenario

-Selected Scenario 3: Developing a desktop application - Develop a simple calculator app for windows and mac

2. Environment planning

Select an IDE: Eclipse

- Java tooling out of box
- Short form for common Java paradigms
- Java debug options
- Includes jpackage (Executable wrapper)

Language: Java

Tools needed:

- Java Swing
 - Cross platform graphics library for window management
 - Decreases development time by having one codebase for all platforms

Executable Wrapper

- Jpackage
 - Required to package as native application files
 - Cross platform packager (Windows, MacOS)
- WiX
 - Required to run jpackage (Windows)
- XCode CLI (MacOS)
 - Required to run jpackage

Hardware Requirements:

- Minimum 8GB ram for both Windows and MacOS
- X86_64 or aarch64 capable cpu

Estimated Setup Time:

Low setup time:

Low Complexity: Download the .exe file if on windows
Download the .app file if on mac

Team Collaboration:

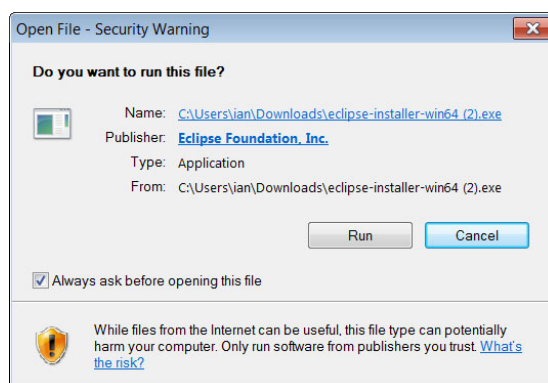
ESTIMATED TIME: 20 MINUTES

Step 1: Install Eclipse

Go to <https://www.eclipse.org/downloads>

Download the latest release of eclipse

Upon opening the file, you will be shown a prompt. Press “Run”



Once it opens, select the “Eclipse IDE for Java developers”

Step 2: Install a JDK

Open the oracle website at [OpenJDK](#)

Download the latest version of OpenJDK

Step 3: Install tools needed for jpackage

Windows:

Install WiX toolset <https://github.com/wixtoolset/wix3/releases>

MacOS:

Open the MacOS terminal

Run the install command in terminal
xcode-select --install

Step 3: Import JavaSwing into your code

Ensure that JavaSwing is imported at the top of your code before packaging it

Step 4: Package your code

Windows

In order to use Jpackage to create the .exe file

Type the command into the CLI:

```
jpackage -t (type) --app-version (version) --icon (/path/to/icon) -n (name) -d (output)
```

Substitute (type), (version), (file path), (name) and (output) with:

(type): Windows "exe" MacOS "dmg"

(version): version of application

(file path): copy the path from your finder

(name): name of application

(output) copy the output file location

HOW TO DOWNLOAD CALCULATOR APP

Windows

Locate the latest release of the .exe file and download it to your device

[ReleasesWindows](#)

MacOS

Locate the latest release of the .dmg file and download it to your device

[ReleasesMacOS](#)